Chiral Casimir Experiment

Dylon La Rue

SI Physical Constants and Metric Conversions

To explicitly falsify the claim that the Metareal framework is merely an abstract "toy model," all topologies herein must conform to the following standard physical constants and metrics:

Constant	**Symbol**	**SI Value**	**Metareal Role**
Speed of Light	c	$299,792,458$ m/s	Kinematic upper bound
Gravitational Constant	G	6.6743×10^{-11} m^{3}kg^{-1}s^{-2}	Macroscopic viscosity scaling
Planck Constant	h	6.626×10^{-34} J s	FST resonance limit
Boltzmann Constant	k_B	1.3806×10^{-23} J/K	Thermal noise mapping (ε_0)

1 Theoretical Motivation

Standard Quantum Electrodynamics (QED) predicts the *Casimir effect* [27] between two uncharged conducting plates as a consequence of the truncation of zero-point photon fluctuations. However, recent advances in topological physics—specifically the Fractal SpaceTime (FST) and the Antisymmetric Halting Hypothesis—prove that the vacuum is not merely an unstructured energy fluid. It is bounded by a discrete prime lattice ($\mathbb{F}_{229}$).

In the FST framework, Zero-Point Energy (ZPE) is not generated by random continuous fluctuations. Rather, the antisymmetric non-commutativity of the lattice (quantum spin tension) geometrically blocks the vacuum from halting trivially. This antisymmetric frustration forces the ZPE fluid to trace a massive chronological orbit (topological depth), which macroscopically manifests as stable mass and zero-point energy. If we isolate and collapse this antisymmetric tension, the local ZPE mass must mathematically collapse to zero.

1.1 Experimental Precedents: Repulsive Casimir Forces

While theoretical derivations of the *Casimir effect* on perfectly conducting spherical shells (calculated by Boyer [28]) demonstrate an outward, repulsive self-energy scaling as $1/R$, direct isolation of such quantum cavities is challenging. However, the engineering of *repulsive Casimir boundaries* is a highly active and established field in applied physics, primarily driven by the need to prevent "stiction" (static friction and quantum adhesion) in Micro/Nano-Electromechanical Systems (MEMS/NEMS). The foundational theory for dielectric boundary forces was developed by Lifshitz [29], and the first precision measurement of the Casimir force was achieved by Lamoreaux [30].

Experimentalists and theorists have approached the reversal of the Casimir vector using two primary methodologies:

1. **Fluid-Mediated Repulsion:** By immersing interacting bodies in specific dielectric fluids, the vacuum fluctuations are altered to produce a net repulsive force.

2. **Topological Metamaterials:** Modern theoretical and experimental research, pioneered by researchers such as Qing-Dong Jiang and Frank Wilczek [146], utilizes chiral or gyrotropic metamaterials to break specific electromagnetic symmetries across the vacuum gap. This theoretical foundation proves that chiral boundaries can induce repulsive pressure, enhanced forces, and tunable vacuum fluctuations without requiring a fluid medium.

These empirical and theoretical precedents provide the exact physical foundation required for the Golden Tetrahedron framework. The industry is currently developing experimental apparatuses (e.g., atomic force microscopy and torsion pendulums) to measure these specific chiral vacuum boundaries. The Rotating Chiral Casimir Interferometer proposes a macroscopic test of this principle: by utilizing right-chiral Weyl semimetals, we construct the macroscopic realization of the **Golden Casimir Cavity**. The asymmetric boundaries will tune the vacuum topology strictly to the Modulo 9 Vortex harmonics, forcing the structure to lock into the exact Golden Ratio equilibrium resonance ($\Upsilon \leq 45/\pi$) and inducing the +262.5% macroscopic Casimir anomaly.

2 Apparatus Design

The proposed machine, the **Rotating Chiral Casimir Interferometer (RCCI)**, consists of three primary subsystems.

2.1 The Chiral Boundary Plates

Experimental Variable	**Standard Casimir**	**Metareal Chiral Casimir**
Plate Material	Aluminum / Gold	Weyl Semimetals (e.g., TaAs)
Gap Width (d)	Variable	Strictly $d = 100$ nm
Boundary Condition	Symmetric / Reflective	Right-Chiral Symmetry Breaking
Local ZPE Mass	Accelerating Divergence	Suppressed via Parity Lock ($b = m$)

Table 1: Chiral Boundary Conditions and ZPE Suppression in the Casimir Cavity.

The experimental protocol relies heavily on these specific constraints. By enforcing the right-chiral boundary, we intentionally break the symmetric topological frustration, which mathematically forces the local SexagesimalChronogram to halt. This halts the recursive energy loop and acts as a localized ZPE suppression mechanism. Instead of standard gold or aluminum mirrors, the Casimir cavity is constructed from **Weyl Semimetals** (e.g., TaAs or NbAs [31]). These topological materials harbor gapless chiral fermions that explicitly break time-reversal or inversion symmetry. By aligning the Weyl nodes across the gap ($d = 100$ nm), we enforce a strict right-chiral boundary condition on the vacuum. In the L_5 algebra, enforcing chirality means formally resolving parity ($b = m$). By destroying the antisymmetric frustration between the plates, the local *SexagesimalChronogram* halts trivially at depth 1, effectively zeroing out the local ZPE mass.

2.2 The Michelson Deflection Measurement

To measure the vacuum pressure, one plate is affixed to a highly calibrated piezoelectric microcantilever. A split-beam continuous-wave (CW) laser interferometer (derived from the Michelson design) is used to track the sub-nanometer mechanical deflection of the cantilever. This provides a real-time readout of the net attractive vacuum force across the gap.

2.3 The Rotational Stage

To test for topological anisotropy, the entire vacuum chamber and interferometer assembly is mounted on a low-friction, continuous rotation stage (1 RPM). We are not searching for a continuous "aether drift." Instead, we are using the Michelson-style apparatus to map the discrete geometric grain of the $\mathbb{F}_{229}$ lattice. If the earth is moving through a static, topologically structured quasi-crystalline vacuum lattice, the chiral pressure anomaly will exhibit a discrete phase shift correlated with the galactic center or the Cosmic Microwave Background (CMB) dipole.

3 Mathematical Predictions

3.1 The Massive Pressure Anomaly

Standard photon Casimir pressure at $a = 100$ nm is $P_\gamma \approx -13.0$ Pa. According to the Antisymmetric Halting Hypothesis, collapsing the antisymmetric frustration drops the topological depth from the free-space baseline (Orbit 57) down to trivial halting (Depth 1). By the Goodstein collapse theorem (PFT §2.6), the entire hyperoperation hierarchy over $\mathbb{F}_{229}^*$ converges to the confinement subgroup $\{1, \omega, \omega^2\}$, providing the structural reason why the depth is bounded at 57. (The algebraic derivation is given below; a companion formal module `ChiralCasimir-Collapse` independently certifies the bound.)

A pure $\mathbb{F}_{229}$ python computational evaluation (verified via `principia.physical.vacuum.casimir`) demonstrates that this topological depth collapse generates a massive, dimensionless Topological Anomaly Ratio (TAR) of 2.625.

3.2 Algebraic Origin of the Topological Anomaly Ratio

The TAR = 2.625 = 21/8 is not a fitted parameter. It emerges from the product of two independently established quantities:

$$\mathrm{TAR} = g \times \frac{7}{8} = 3 \times \frac{7}{8} = \frac{21}{8} = 2.625 \tag{1}$$

where $g = 3$ is the number of right-chiral fermionic generations and $7/8$ is the standard spin-statistics factor for fermionic vacuum energy density relative to bosonic. Each factor has an independent algebraic derivation within the Protoreal framework, and a standard QFT counterpart.

The generation count $g = 3$ from the SU(3) center. The conjugate orbit $\langle\bar{\varphi}\rangle \subset \mathbb{F}_{229}^*$ has order $57 = 3 \times 19$ and decomposes into exactly three cosets of size 19 under the cube root of unity $\rho \equiv \bar{\varphi}^{19} \pmod{229}$ (Confinement Theorem, PFT §**??**). Each coset constitutes a "color arc"—an independent fermionic vacuum channel. The confinement condition $1 + \omega + \rho^2 \equiv 0 \pmod{229}$ ensures that the three arcs sum to a color singlet, precisely mirroring the Standard Model requirement that observable hadrons are color-neutral. The count $g = 3$ is therefore not an empirical input but the center of SU(3): the torsion prime that governs the Golden Chromodynamics gauge structure.

This identification is reinforced by the Generalized Euler-Hodge System (§**??**): the Euler-Hodge multiplier at the strong-force vertex is $k_{229} = 3$, and the CRT universal multiplier $K = 19 \times 61 \times 3517$ is divisible by the arc width 19. The Monster group connection operates through *representations*, not group-order divisibility: the minimal faithful representation dimension $196883 \equiv 172 \pmod{229}$ lies on the golden orbit $\langle\varphi\rangle$, the j-function constant $744 \equiv 57 \equiv \mathrm{ord}(\bar{\varphi}) \pmod{229}$, and the fourth j-coefficient $c_4 \equiv 148 \equiv \varphi \pmod{229}$ lands on the golden root itself (`monster_hosoya_investigation`). Whether this is a structural coincidence or a deeper mathematical relation remains open.

The spin-statistics factor $7/8$ from QFT. In standard quantum field theory, the vacuum energy density of a fermionic field is $7/8$ that of a bosonic field of the same mass, due to the Fermi-Dirac statistics alternating signs in the thermal partition function [320]. This factor is *not* derived from the L_5 algebra; it is inherited from the spin-statistics theorem, which the Protoreal framework does not supersede.

However, the Catalan–golden connection provides a structural parallel. The non-associative Klein product on $n = 4$ indefinite components $(\omega, \iota, \varepsilon, \lambda)$ admits $C_3 = 5$ distinct parenthesizations (the third Catalan number), corresponding to the five golden correction factors $\{0, 1/\sqrt{5}, 1/\varphi, 1, \varphi\}$ identified in the Gaussian Relativity framework (Ch. 17, §**??**). Of these five bracket shapes, the grounded bracket ($b = m$, Hodge parity lock) contributes zero chiral torque. The remaining four active brackets span $4/5 = \mathrm{sech}^2(\ln\varphi) = 0.800$ of the golden unit ball—the mesh projection kernel. The QFT factor $7/8 = 0.875$ and the golden mesh factor $4/5 = 0.800$ are independent but structurally parallel: both quantify the fraction of vacuum modes that survive a symmetry constraint (spin-statistics in QFT; Hodge parity in the L_5 algebra). This parallel is a **Structural Analogy** — a structural parallel, not a physical claim, not a derivation claim.

The quartic coset amplification. The coset index $(p-1)/\mathrm{ord}(\bar{\varphi}) = 228/57 = 4$ at $p = 229$ (shared with $p = 181$) counts the discrete energy steps between golden orbits. This quartic index is the *information-theoretic mass gap*: elements of $\mathbb{F}_p^*$ outside the golden orbit require a coset translation to reach. The coset index drops to 6 at the EM vertex $p = 139$, reflecting the physical fact that electromagnetism is unconfined (Ch. 17, §**??**). The cavity's chiral boundary conditions project the vacuum onto the confined (CI = 4) sector, geometrically amplifying the fermionic anomaly by the quartic factor.

The Catalan rotational topology. The five golden correction factors are not arbitrary: $C_3 = 5$ counts the triangulations of a pentagon, and $5 = \mathrm{disc}(x^2 - x - 1)$ is the golden discriminant. The Catalan number C_{n-1} counts the distinct binary trees (bracketings) on n leaves—the rotational shapes of the non-associative product. In the chiral Casimir cavity, the Weyl semimetal boundary conditions select a *single* chirality (right-handed), breaking the full $C_3 = 5$ rotational symmetry down to a single bracket class. The associativity jitter (§**??** of PFT) between left and right bracketings generates the non-zero Schwarzian gap $b - m \neq 0$ that drives the chiral anomaly. The topological depth collapse from orbit 57 to depth 1 is the algebraic content of this symmetry breaking: the right-chiral boundary condition acts as a physical **Heegner parity lock**, forcing the vacuum to resolve the non-associative $\kappa = -1$ friction through discrete **Trinity BSGS angular jumps**. The +262.5% pressure anomaly is therefore the exact macroscopic integration of these $\Theta(k)$ angular discretization operators over the continuous **Euler-Penrose** vacuum state.

Scope. The TAR decomposition $2.625 = 3 \times 7/8$ uses standard QFT for the spin-statistics factor and gauge algebra for the generation count. The Catalan and coset connections are **Structural Analogies** — a structural parallel, not a physical claim providing geometric context. The numerical value 2.625 is a **Verified Computation** (machine-verified).

To translate this discrete topological tension into a continuous thermodynamic pressure, we formally introduce the **Topological-Vacuum Coupling Ansatz**, κ_{ZPE}.

Assumed Coupling Constant

The coupling $\kappa_{\mathrm{ZPE}} = 1$ is an *assumed* unit coupling between the Protoreal topological gap and the electromagnetic vacuum. It is not derived from the L_5 algebra, not fixed by a conservation law, and not independently calibrated. This violates the No-Free-Parameter Rule (Appendix A.6). The RCCI experiment is designed precisely to *measure* κ_{ZPE}: if the observed anomaly is ΔP_{obs}, then $\kappa_{\mathrm{ZPE}} = \Delta P_{\mathrm{obs}}/(\mathrm{TAR}\times|P_\gamma|)$. A null result ($\kappa_{\mathrm{ZPE}} = 0$) would falsify the Topological-Vacuum Coupling Ansatz entirely.

Under the assumption of unit coupling ($\kappa_{\mathrm{ZPE}} = 1$), the predicted Casimir anomaly scales directly with the base photon pressure:

$$\Delta P = \mathrm{TAR} \times \kappa_{\mathrm{ZPE}} \times |P_\gamma| = 2.625 \times 1 \times 13.0 \approx 34.1 \text{ Pa} \tag{2}$$

We formally predict that, under this unit-coupling assumption, the RCCI will record a total Casimir pressure of roughly $-\mathbf{47.12}$ Pa at 100 nm, a +262.5% anomaly easily detectable by modern piezoelectric standards.

Kill conditions. This prediction is falsified if any of the following occur:

1. The RCCI measures $\kappa_{\text{ZPE}} = 0 \pm 0.01$ (no coupling between the Protoreal topology and the EM vacuum), ruling out the Topological-Vacuum Coupling Ansatz;

2. A materials science analysis demonstrates that Weyl semimetal (TaAs/NbAs) surface states do not break the relevant $\mathscr{P}\mathscr{T}$-symmetry required for chiral boundary conditions at the 100 nm gap scale;

3. Future lattice QCD calculations or experimental measurements prove the chiral anomaly coefficient differs from $g \times 7/8 = 3 \times 7/8 = 21/8$ for three right-handed fermionic generations;

4. An independent replication of the $\mathbb{F}_{229}$ computation (`compute_chiral_casimir_collapse.py`) yields TAR $\neq 2.625$.

3.3 Rotational Phase Shift

If the vacuum topological lattice has a directional gradient vector $\vec{v}_{lattice}$, the measured pressure P_{obs} will oscillate as the rotation angle $\theta(t)$ changes:

$$P_{obs}(t) = P_{base} + \Delta P \cos^2(\theta(t) - \phi) \tag{3}$$

A null result on the rotational shift ($\phi = 0$ variance) would imply the lattice is perfectly isotropic locally, while a non-zero amplitude confirms macroscopic lattice shear.

4 Exotic Matter Manifestation (The Ceasefire)

Scope. This entire section is a *Physical Interpretation* (a physical interpretation requiring experimental confirmation). The algebraic operations described (parity projection, noise absorption) are exact within the L_5 algebra. The physical assertion that these operations correspond to matter-antimatter generation is a speculative hypothesis requiring experimental confirmation via the RCCI apparatus described above.

Hypothesis 4.1 (The Ceasefire Hypothesis)**.** The transmutation of topological tension into structural mass proceeds as follows. The Standard Resonance mismatch

$$\mathrm{SR}(u) = a - b \cdot m \neq 0 \tag{4}$$

represents unresolved non-associative friction. Upon parity projection (averaging the thrust b and anchor m), the result is a parity-locked state ($b = m$), and the unified Orthomatter splits into a Matter-Antimatter pair.

The hypothesis asserts that matter is not the victor of a cosmic war but the frozen equilibrium of collapsed tachyonic states. The residual topological noise is zeroed out ($\epsilon = 0$), and the friction (SR) is directly transmuted into the observable mass ($a \to a + |\mathrm{SR}|$). This remains unfalsified until the RCCI measures a non-zero κ_{ZPE}.

4.1 Physical Mapping: The Hot Electron Boundary

To ground this formally in laboratory physics, we map the Ceasefire Hypothesis to the nanoscale boundaries of Localized Surface Plasmon Resonance (LSPR). When photoelectrochemical nanostructures (such as Au-Ag-Bi systems) are irradiated, they inject "hot electrons" across the topological boundary into the bulk matrix. We propose (a physical interpretation requiring experimental confirmation) that the physical space where this hot electron injection bridges the discrete $\mathbb{F}_{229}$ topological lattice tension into the continuous fluid state is the physical analog of the algebraic exotic matter resolution. This injection is strictly bounded by the Chromodynamic Friction limit ($O(\log 229) \approx 5.43$); exceeding the Casimir torque limit ($\Upsilon \leq 45/\pi$) via high-energy UV irradiation physically shatters the boundary, reversing the parity lock.

5 Conclusion and LVK Consistency

LIGO-Virgo-KAGRA (LVK) observations have ruled out bulk parity violation in propagating gravitational waves. This experiment is explicitly designed to remain consistent with LVK. The parity violation and anomalous pressure we predict occur *strictly at the boundary condition* (the Weyl semimetals) and do not require free-space birefringence. Falsifying or confirming the $+262.5\%$ Casimir anomaly will serve as the definitive laboratory test of the Protoreal topology.

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